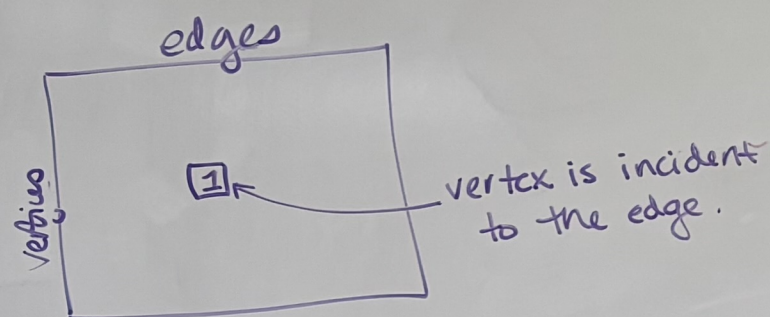


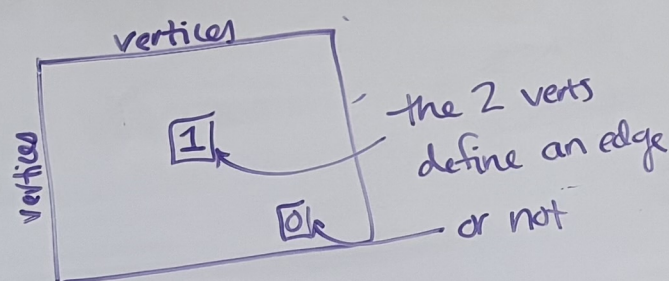
Handout 3: Union Find Algo.

Data Structures for Graphs (Edges specifically)

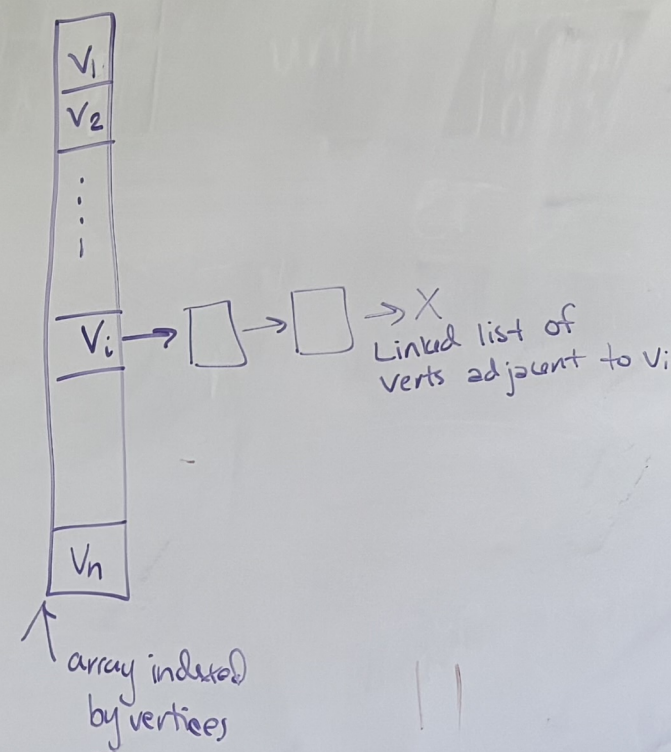
① Incidence Matrix (2d array)



② Adjacency Matrix (2D array)



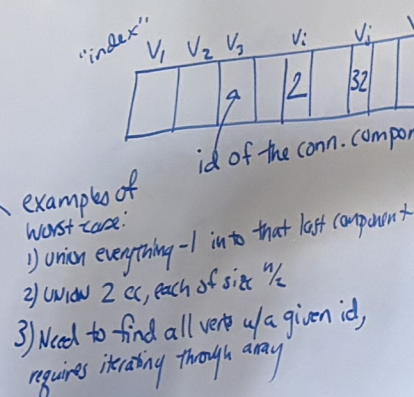
③ "Sparse Matrix": Adjacency List (an array of lists)



④ Bag of Edges - Yikes!

UNION-FIND ALGO

- S = stores the connected components, starts w/ 1 vertex/component, then "adds" one edge at a time.
- UNION: merge 2 connected components
- FIND: finds the conn. component containing a given vertex
- Quick Find:
 - 1 UNION is $\Theta(n)$
 - 1 FIND is $\Theta(1)$
- Quick Union:
 - 1 UNION is $\Theta(1)$ assuming we know the roots
 - 1 FIND is $\Theta(n)$
- w/ Heuristics: m operations is



UNION-FIND Algo

- S = stores the connected components, starts w/ 1 vertex / component, then "adds" one edge at a time.

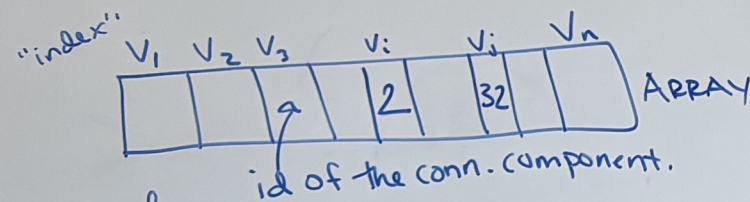
- UNION: merge 2 connected components

- FIND: finds the conn. component containing a given vertex

- Quick Find:
1 UNION is $\Theta(n)$
1 FIND is $\Theta(1)$

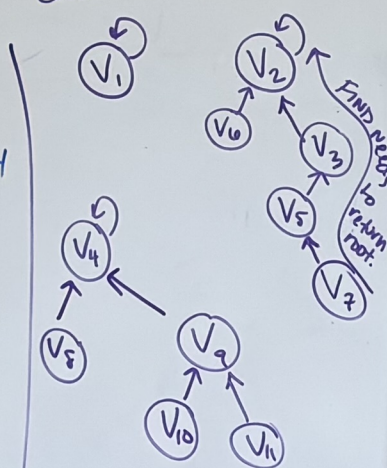
- Quick Union:
1 UNION is $\Theta(1)$ assuming we know the roots
1 FIND is $\Theta(n)$

- w/ Heuristics:
m operations is $\Theta(m \alpha(n))$
 $\uparrow$ INVERSE ackermann fn.



- examples of worst case:
- 1) union everything - 1 into that last component
 - 2) union 2 cc, each of size $n/2$
 - 3) Need to find all vert w/ a given id, requires iterating through array

QUICK UNION CARTOON:



$$S = \{V_1, V_2, \dots, V_n\}$$

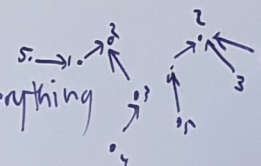
$\downarrow$ UNION (V_2, V_3)

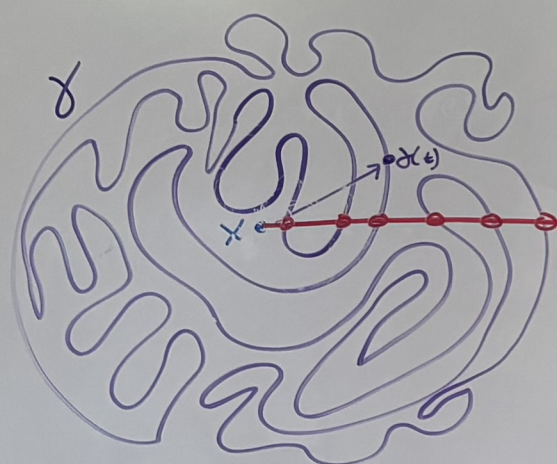
$$S = \{V_1, \{V_2, V_3\}, V_4, V_5, \dots, V_n\}$$

UF Heuristics:

- 1) Depth Control
add a variable called 'rank'
→ always repoint "shorter to taller"
→ if nothing else done, rank = depth

- 2) Fatten Up
→ when I find, repoint everything I touch to the root.





Ans: YES iff $\frac{\gamma(t)-x}{\|\gamma(t)-x\|}$ goes one time around S^1 .

γ = simple, closed curve in $\mathbb{R}^2$
 $\gamma: S^1 \rightarrow \mathbb{R}^2$ "nice" embedding
 $\bullet x \in \mathbb{R}^2$

IS x inside or outside of γ ?

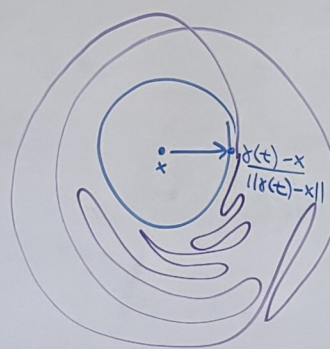
Jordan Curve Thm: A simple, closed curve in $\mathbb{R}^2$ creates 2 conn. components: ① homeomorphic to $\mathbb{D}^2 := \{(x,y) \in \mathbb{R}^2 \mid x^2+y^2 \leq 1\}$ ② infinite "outside"

WINDING

$\gamma: S^1 \rightarrow \mathbb{R}^2$ "nice" immersion

The winding # is how many times

$\frac{\gamma(t)-x}{\|\gamma(t)-x\|}$ wraps around S^1

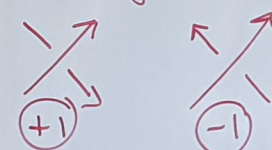


Inside/outside γ

- Draw a ray (generic/non-tangent)
- Count how many times it crosses
 even $\Rightarrow$ outside
 odd $\Rightarrow$ inside

Winding

- Orient the curve
- Draw a ray
- Count Signed Intersections



- Sum = the winding number!
 (depends on γ + choice of direction)